

Tools

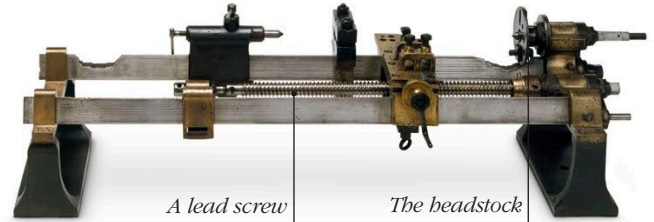
The onset of the Industrial Revolution in the 1760s required a new set of tools beyond the basic hammers and chisels of old. The scale and volume of production, using new materials including metals, now required more power, more speed, and more precision than a human alone could achieve.

CIRCULAR SAW

Traditionally, logs were sawed by hand using a pit saw, with an up-down motion. This was slow and inefficient. In 1813, Tabitha Babbitt, a member of a Shaker community in the US, introduced the first circular saw in a saw mill, which was powered by water.

SCREW-CUTTING LATHE

A lathe spins metal against a tool to give it a circular shape or cut a screw thread. This could be done by hand but, in the 1790s, the Englishman Henry Maudslay and American David Wilkinson invented lathes separately, in which the tool was driven by a screw geared to the lathe.



A lead screw moves the tool as the workpiece rotates.

The headstock grips and rotates the workpiece.

Timber is pushed through a circular saw that requires more power to operate than a hand saw, but cuts much faster.

